

Second-Hand Vehicle Marketplace with Intelligent Search and Automated Inventory Processing

PID = 11

Concept:

A B2B platform where car dealers upload their vehicle inventory in bulk, the system processes and enriches each listing through an automated pipeline, and buyers can search the catalog using either traditional filters or plain natural language queries.

Background

The automotive resale market is increasingly driven by digital inventories, yet small to medium-sized car dealerships often struggle with the technical overhead of managing digital catalogs. Currently, many dealers manually enter vehicle data across various platforms, leading to inconsistencies, slow updates, and poor data quality. Furthermore, traditional search interfaces on B2B platforms rely heavily on rigid filtering systems (e.g., Year, Make, Model), which can be restrictive for buyers looking for specific features or value-based outcomes. As the industry moves toward cloud-native architectures, there is a growing need for platforms that not only centralize inventory but also leverage automation to "enrich" data by automatically identifying features or correcting specifications, to improve the buying experience.

Main Focus:

The project emphasizes cloud-native architecture and engineering best practices over feature depth or frontend complexity. Core areas of focus are asynchronous event-driven processing, infrastructure as code, a fully automated CI/CD pipeline with testing at multiple levels, and the ability to scale horizontally under load. The system runs identically in a local Docker environment and on AWS, with no (or less) code changes required between environments.

Problem Description

Car dealers face significant bottlenecks when uploading large volumes of vehicle data. Bulk uploads often result in "dirty data" with missing specifications, non-standardized formatting, and poor SEO visibility. On the buyer side, navigating vast inventories using only standard filters is often inefficient; a buyer looking for "a family-friendly SUV under \$30k with leather seats" must navigate multiple menus rather than simply stating their need. Existing enterprise-grade solutions offer bulk processing and advanced search but are often prohibitively expensive and lack the portability of modern cloud-native designs. There is a need for a streamlined B2B platform that combines automated data enrichment with an intuitive, natural language-driven search interface, built on a robust, portable infrastructure.

Proposed Solution

This project proposes a cloud-native B2B platform designed for the automotive sector. The system allows dealers to upload vehicle inventory in bulk (via CSV/JSON), which then triggers an Automated Enrichment Pipeline. This pipeline processes the data to standardize attributes and enhance listing quality.

For buyers, the platform provides a dual-mode search experience: a Traditional Filter-Based Search and a Natural Language Processing (NLP) Search, allowing users to query the catalog using plain English. To ensure modern engineering standards, the entire stack is developed to be "environment agnostic," running identically on local Docker environments and AWS. The lifecycle of the application from code commit to deployment is managed via a fully automated CI/CD pipeline.

Objectives

The main objectives of this project are:

- To design a scalable B2B architecture for bulk data ingestion and processing.
- To implement an automated pipeline that enriches vehicle listings with standardized metadata.
- To develop a hybrid search engine supporting both structured filters and natural language queries.
- To build a cloud-native infrastructure using AWS services while maintaining local parity via Docker.
- To establish a robust CI/CD pipeline for automated testing and deployment.

Main Functionalities:

1. Dealer Inventory Management: Dealers can upload vehicle listings in bulk including photos and structured data (make, model, year, mileage, fuel type, gearbox, color, price, condition). Each listing has a unique identifier and belongs to a dealer account.
2. Automated Listing Pipeline: Uploaded inventory goes through an automated processing pipeline covering validation of required fields, image processing into multiple resolutions, and metadata normalization to ensure consistent data quality across all dealers.
3. Vehicle Search and Filtering: Buyers can browse the catalog and filter by make, model, year range, price range, mileage, fuel type, gearbox type, color, and location. Results are paginated and sortable.
4. Natural Language Search: Buyers can describe what they are looking for in plain text such as "automatic diesel SUV under 20 million LKR with less than 80000km" and the system translates this into a structured search query automatically.
5. Pipeline Status and Notifications: Dealers receive automated email notifications when their inventory upload has been processed successfully, including a summary of imported listings and any entries that were rejected and the reason why.
6. Inventory Dashboard: Dealers can view and manage their active listings, see which ones are live in the catalog, and track the status of any pending uploads.
7. Synthetic Data Generation: The system includes tooling to generate realistic vehicle listing datasets at scale for testing and development purposes.

Scope of the Project

The project covers the end-to-end development of a full-stack web application, cloud infrastructure, and DevOps automation. **(following are possible consideration. Feel free to modify as needed, based on the defined requirement)**

Frontend (Web Application)

- Dealer dashboard: interface for bulk file uploads, inventory management, and status tracking of the enrichment process.
- Buyer catalog: high-performance search interface featuring a "Smart Search" bar for natural language and a sidebar for traditional faceted filters.
- Responsive design: optimized for both desktop and mobile browsers.

Backend services (Full-Stack)

- Ingestion engine: handles bulk uploads and validates data integrity.
- Enrichment pipeline: background processing service (potentially using AWS Lambda or worker containers) to clean and standardize vehicle data.
- Search service: integration with a search-optimized database to handle complex queries and NLP.
- Authentication: secure multi-tenant access for dealers and buyers.

Cloud & DevOps (Infrastructure)

- Cloud-Native AWS: Utilization of S3 for storage, RDS/DynamoDB for data, and ECS/EKS for orchestration.
- Local parity: full containerization using Docker and Docker Compose to ensure the "runs on my machine" experience matches production.
- CI/CD pipeline: automated workflows (e.g., GitHub Actions or GitLab CI) for linting, unit/integration testing, and automated deployment to AWS.

Out of Scope

- Integration with physical vehicle inspection hardware.
- Real-time financial transaction processing/escrow services.
- Advanced Generative AI for virtual car showrooms.

Expected Outcomes

By the end of the project, the student group is expected to deliver:

- Functional B2B Platform: live web application demonstrating bulk upload and intelligent search.
- Enrichment logic: documented pipeline showing how raw data is transformed into enriched listings.
- Infrastructure as Code (IaC): Scripts (e.g., Terraform or CloudFormation) and Dockerfiles used to deploy the system.
- CI/CD report: documentation of the automated testing and deployment stages.
- Technical report: comprehensive analysis of the system architecture, cloud-native decisions, and performance evaluation.